Extra Dessert #9: Television Practices (ATV and SSTV)

Fast-scan TV (NTSC and DVB-T) standards, slow-scan TV (SSTV) encoding, vestigial sideband, VIS codes.

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Fast-Scan TV (NTSC)

Question	Answer	Q
NTSC horizontal lines per frame	**525**	E2B02
How interlaced scanning is generated	**By scanning odd-numbered lines in one field and even-numbered lines in the next**	E2B03
Vestigial sideband modulation	**AM in which one complete sideband and a portion of the other are transmitted**	E2B06
Why use VSB in analog fast-scan TV	**Reduces the bandwidth while increasing the fidelity of low frequency video components**	E2B05
Use commercial analog TV receivers on 70 cm	**Transmitting on channels shared with cable TV**	E2B08

Digital TV (DVB-T)

Question	Answer	Q
Coding rate of 3/4 means	**25% of the data sent is forward error correction data** (1 in every 4 bits is overhead for error correction)	E2B01
Modulation types used for amateur DVB-T	**QAM and QPSK**	E2B07

Slow-Scan TV (SSTV)

Question	Answer	Q
Color information in analog SSTV	**Color lines are sent sequentially** (order varies by mode – R/G/B in Scottie/Martin, Y/Ry/By in Robot Color)	E2B04
What encodes brightness	**Tone frequency**	E2B10
Function of the VIS code	**To identify the SSTV mode being used**	E2B11
What signals SSTV software to begin a new line	**Specific tone frequencies** (sync tones)	E2B12
Receiver type for DRM-protocol SSTV	**SSB** (DRM is sent within an SSB-bandwidth channel)	E2B09

Dessert 9 Question Bank (E2B)

E2B01. In digital television, what does a coding rate of 3/4 mean?
A. 25% of the data sent is forward error correction data.
B. Data compression reduces data rate by 3/4.
C. 1/4 of the time interval is used as a guard interval.
D. Three, four-bit words are used to transmit each pixel.

E2B02. How many horizontal lines make up a fast-scan (NTSC) television frame?
A. 30.
B. 60.
C. 525.
D. 1080.

E2B03. How is an interlaced scanning pattern generated in a fast-scan (NTSC) television system?
A. By scanning two fields simultaneously.
B. By scanning each field from bottom-to-top.
C. By scanning lines from left-to-right in one field and right-to-left in the next.
D. By scanning odd-numbered lines in one field and even-numbered lines in the next.

E2B04. How is color information sent in analog SSTV?
A. Color lines are sent sequentially.
B. Color information is sent on a 2.8 kHz subcarrier.
C. Color is sent in a color burst at the end of each line.
D. Color is amplitude modulated on the frequency modulated intensity signal.

E2B05. Which of the following describes the use of vestigial sideband in analog fast-scan TV transmissions?
A. The vestigial sideband carries the audio information.
B. The vestigial sideband contains chroma information.
C. Vestigial sideband reduces the bandwidth while increasing the fidelity of low frequency video components.
D. Vestigial sideband provides high frequency emphasis to sharpen the picture.

E2B06. What is vestigial sideband modulation?
A. Amplitude modulation in which one complete sideband and a portion of the other are transmitted.
B. A type of modulation in which one sideband is inverted.
C. Narrow-band FM modulation achieved by filtering one sideband from the audio before frequency modulating the carrier.
D. Spread spectrum modulation achieved by applying FM modulation following single sideband amplitude modulation.

E2B07. Which types of modulation are used for amateur television DVB-T signals?
A. FM and FSK.
B. QAM and QPSK.
C. AM and OOK.
D. All these choices are correct.

E2B08. What technique allows commercial analog TV receivers to be used for fast-scan TV operations on the 70-centimeter band?
A. Transmitting on channels shared with cable TV.
B. Using converted satellite TV dishes.
C. Transmitting on the abandoned TV channel 2.
D. Using USB and demodulating the signal with a computer sound card.

E2B09. What kind of receiver can be used to receive and decode SSTV using the Digital Radio Mondiale (DRM) protocol?
A. CDMA.
B. AREDN.
C. AM.
D. SSB.

E2B10. What aspect of an analog slow-scan television signal encodes the brightness of the picture?
A. Tone frequency.
B. Tone amplitude.
C. Sync amplitude.
D. Sync frequency.

E2B11. What is the function of the vertical interval signaling (VIS) code sent as part of an SSTV transmission?
A. To lock the color burst oscillator in color SSTV images.
B. To identify the SSTV mode being used.
C. To provide vertical synchronization.
D. To identify the call sign of the station transmitting.

E2B12. What signals SSTV receiving software to begin a new picture line?
A. Specific tone frequencies.
B. Elapsed time.
C. Specific tone amplitudes.
D. A two-tone signal.

Answer Key

E2B01: A	E2B02: C	E2B03: D	E2B04: A
E2B05: C	E2B06: A	E2B07: B	E2B08: A
E2B09: D	E2B10: A	E2B11: B	E2B12: A